

Jason S. Lucas

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Summary

AI safety and trustworthy-AI researcher with ten years of interdisciplinary experience across NLP, cybersecurity, health informatics, and epidemiology. Founding Director of the Secure and Ethical AI Lab (SEAL) at CU Boulder, with a research program focused on multilingual NLP, low-resource AI, dialectal robustness, and equitable language technologies. Published at top NLP and AI venues (ACL, EMNLP, NAACL, ICML, KDD, IEEE) with translational collaborations at Lawrence Livermore National Lab, SoundHound, Amazon, and Coalfire.

Experience

Assistant Professor and Director, SEAL Lab Aug. 2026 – Present
University of Colorado Boulder, Department of Information Science (CMDI)

Senior Researcher, JSALT 2026 Workshop Jun. 2026 – Aug. 2026
Johns Hopkins University, Center for Language and Speech Processing

- Selected for invitation-only 12th Frederick Jelinek Memorial Workshop; leading research on uncertainty-aware LLM-based health coaching agents with human oversight for sustainable behavior change.

Doctoral Researcher, Amazon Nova AI Challenge (Team Lion-0xA) Feb. 2026 – Present
The Pennsylvania State University & Amazon

- One of 10 global finalist teams from hundreds of applicants worldwide (\$250K team sponsorship); red-team researcher probing agentic AI systems for adversarial vulnerabilities in multi-step software-engineering tasks.

Graduate Research Assistant, PIKE Lab Aug. 2021 – May 2026
The Pennsylvania State University, College of IST (Advisor: Dr. Dongwon Lee)

- Led multilingual NLP research spanning 70+ languages on disinformation detection, harmful-content classification, and AI safety guardrails; authored 14+ peer-reviewed papers (ACL, EMNLP, NAACL, ICML, AAAI, IEEE Computer) with 315+ citations.
- Built methods for dialectal evasion attacks and defenses against safety-aligned LLMs (DIA-HARM, DIA-Guard, SHIELD, BLUFF benchmarks); co-authored 3 NSF and NIH grant proposals; mentored 8+ research mentees.

DSSI Graduate Research Intern May 2025 – Aug. 2025
Lawrence Livermore National Laboratory, Data Science Institute

- Conducted secure-AI research on Mixture-of-Agents and Mixture-of-Experts architectures and cross-lingual bridge models; evaluated LLM vulnerabilities for national-security-relevant multilingual and multimodal AI.

NLP Research Intern, SoundHound AI (Interactions LLC) May 2024 – Aug. 2024
Developed the Chain-of-Interactions framework for iterative ICL on conversational-AI dialogues; published at EMNLP 2025.

Cybersecurity Consultant Intern, Coalfire Systems May 2023 – Aug. 2023
PCI DSS compliance assessments; encryption verification with Wireshark and browser DevTools.

Adjunct Assistant Professor & Clinical Skills Lead, St. George's University Aug. 2010 – Jul. 2021
Taught IT, Medicine, and Public Health courses; led the B-Line Division and SimCapture clinical-simulation platform across the School of Medicine.

Education

Ph.D., Informatics (Data Science & AI/NLP)	The Pennsylvania State University	2026
M.P.H., Epidemiology	St. George's University, Grenada	2020
M.Sc., Computer Information Systems	Boston University, Health Informatics concentration	2014
B.Sc., Information Technology	St. George's University, Grenada (<i>Cum Laude</i>)	2010

Selected Publications

1. **J. Lucas**, P. Irikefe, A. Uchendu, U. Najaer, C. Adejoro, P. Sterling, and D. Lee. Position: Breaking the Dual Curse of Multilingual AI Requires Socio-Technical Guardrails. *ICML 2026*.
2. **J. Lucas**, M. Murtagh, A. Al-Lawati, A. Uchendu, and D. Lee. DIA-HARM: Dialectal Disparities in Harmful Content Detection across 50 English Dialects. *ACL 2026 (Social Impact Award)*.
3. **J. Lucas**, J. Chen, A. Al-Lawati, M. Nahar, and M. Mehrabani. Chain-of-Interactions: Iterative ICL for Abstractive Task-Oriented Dialogue Summarization. *Findings of EMNLP 2025*.
4. E. Artemova, **J. Lucas**, S. Venkatraman, J. Lee, S. Tilga, A. Uchendu, and V. Mikhailov. Beemo: Benchmark of Expert-Edited Machine-Generated Outputs. *NAACL 2025*.
5. **J. Lucas**, A. Uchendu, M. Yamashita, J. Lee, S. Rohatgi, and D. Lee. Fighting Fire with Fire: The Dual Role of LLMs in Crafting and Detecting Elusive Disinformation. *EMNLP 2023*.
6. **J. Lucas**, B. Maung Maung, M. Tabar, K. McBride, and D. Lee. The Long-Tail Impact of Generative AI on Disinformation. *IEEE Intelligent Systems*, 2024.

Selected Datasets and Benchmarks

DIA-HARM	50-dialect harmful content benchmark	https://github.com/js15710/dia-harm
SHIELD	Safety evaluation benchmark for dialectal evasion attacks //huggingface.co/datasets/js15710/Shield	https://huggingface.co/datasets/js15710/Shield
BLUFF	Falsehoods in long-tail multilingual low-resource languages //huggingface.co/datasets/js15710/BLUFF	https://huggingface.co/datasets/js15710/BLUFF
DIA-Guard	AI safety evaluation and defense framework	https://github.com/js15710/dia-guard

Selected Awards & Distinctions

Social Impact Paper Award, Association of Computational Linguistics, Main Conference San Diego	2026
Ph.D. Award for Research Excellence, Penn State College of IST	2026
NSF NRT Graduate Research Fellowship in Linguistic Diversity (LinDiv)	2022 – 2024
ICML & Black in AI Diversity & Inclusion Scholarship	2022
ACL Diversity & Inclusion Scholarship	2022
Pan-African Professional Alliance Leadership Award, Penn State	2024

Technical Skills

Programming Languages	Python, Java, SQL, LaTeX, HTML5, CSS, PHP, R, C++
Machine Learning	PyTorch, TensorFlow, Hugging Face Transformers, Adapter-Transformers, AdapterHub, vLLM, Pandas, NumPy, scikit-learn
Platforms & Tools	VS Code, PyCharm, IntelliJ IDEA, Google Cloud, Microsoft Azure, Streamlit, Twitter API, Selenium, Tesseract
Domain Expertise	Multilingual NLP, AI Safety, Adversarial ML, Health Informatics, Epidemiology, EHR Systems

Languages

English (Native), Grenadian English Creole (Native)

Research Links

GitHub	https://github.com/js15710
Google Scholar	https://scholar.google.com/citations?user=XU1WN6YAAAAJ
Semantic Scholar	https://www.semanticscholar.org/author/Jason-Samuel-Lucas/2260072705
Homepage	https://www.jasonslucas.com